Part 2: TPUs

(Q1) 200B parameters bf16 split over 32 TPUv4p

400B bytes = 4.00e11 bytes = (4.00e11 / 32) bytes per chip

TPUv4p HBM BW = 1.2e12 bytes/second

$$t_{comms\ per\ chip} = \frac{4.00e11\ bytes}{32 \times 1.2e12\ bytes/second} = 1.04e^{-2}\ seconds$$

1ms = 1e-3 sec

1e-2 s = 10 ms

(Q2) Full TPU v5e pod

Total CPU hosts = num chips / chips per host = (16 × 16) / (4 × 2)

Num TPU processor cores = chips × 1 for v5e = 256 = 256 / 5 = 52

HBM = 16 × 256 = 4TB

Total FLOPs/s = chips × FLOPs/s per chip = 256 × 1.97e14

Total HBM BW/chip = 256 × 8.2e11 = 2.10e14 = 5.04e16 FLOPs/s (bf16)

Full TPU v5p pod

HBM = 8960 × 96 GB = 860TB

Total Hosts = $\frac{16 \times 20 \times 28}{4 \times 2 \times 1} = 2240$ hosts

Num TPU cores = 16 × 20 × 28 × 2 = 17920 cores

Total FLOPs/s = (16 × 20 × 28) × 4.59e14 = 4.11e18

Total HBM BW/chip = 8960 × 2.8e12 = 2.51e16

(Q3) PCIe BW = 1.6e10 bytes/second , A is bf16[D, F]

X is bf16[B, D]

let B << D and F = 4D

Operating on TPU v5e, so Intensity(accelerator) = $\frac{9.20e14 \text{ FLOPs/s}}{1.6e10 \text{ bytes/s}} = 57{,}500$

(since t_{comms} will be limited by PCIe)

$$\text{Intensity(matmul)} = \frac{BF(2D-1)}{2BD + 2DF + 2BF} = \frac{B(4D)(2D-1)}{2BD + 2DF + 2BF}$$

$$= \frac{B(4D)(2D-1)}{2(D)(4D)} = \frac{2B}{2} = B$$

for B ≥ 57,500 , matmul becomes compute bound

(Q4) 1 TPU v5e

int8[B, 4096] · int8[4096, 16384] , where B is unknown

1. How long will this take?

⟶ int OPs needed = $B(2(4096)-1)(16384) = 1.34e8 B$ OPs

⟶ bytes to load = $4096B + (16384)(4096) + 16384B$

$= 1.34e8 + 2.04e4 B$ bytes

so $t_{matmul} = \frac{1.34e8 B}{1.34e8 + 2.04e5 B}$

as B gets larger

$$t_{comms} = \frac{1.34e8 + 2.04e5\,B \text{ bytes}}{8.2e11 \text{ bytes/second}}$$

$$t_{math} = \frac{1.34e8 B \text{ ops}}{3.94e14 \text{ FLOPs/s}}$$

$t_{comms} = 1.6e\text{-}4 \text{ seconds} + 2.49e\text{-}7\,B$

$t_{math} = 3.40e\text{-}7\,B$

$t_{matmul} = \max(t_{comms}, t_{math})$

VMEM takes about 22x less time than HBM. So

$t_{comms,vmem} = \frac{t_{comms,HBM}}{22} = \frac{1.4e^{-4} + 2.49e^{-7}B}{22}$,

then take $\max(t_{comms,vmem}, t_{math})$

Q4) TPUv5e:

int8 [B, 4096] · int8 [4096, 16384]

$t_{math} = \frac{\text{total int ops}}{3.94e14 \text{ ops/s}} = \frac{2(B)(4096)(16384)}{3.94e14} = \frac{1.34e8B}{3.94e14}$ s

$t_{comms} = \frac{\text{total bytes/written bytes}}{8.2e11 \text{ bytes/second}} = \frac{(4096)(16384) + B(4096) + B(16384)}{8.2e11}$ s

$t_{max} = \max(t_{math}, t_{comms})$.

Intensity (accelerator) $= \frac{3.94e14}{8.2e11} = 480$. So we want

$$\frac{1.34e8B}{(4096)(16384) + 4096B + 16384B} \geq 480$$

$$1.34e8B - (480)4096B - (480)(16384B) \geq (480)(16384)(4096)$$

$$= B(1.34e8 - 2e6 - 8e6) \geq 32e9$$

$$= 1.24e8B \geq 32e9, \text{ so } B \geq \frac{32e9}{1.24e8} \approx 258$$

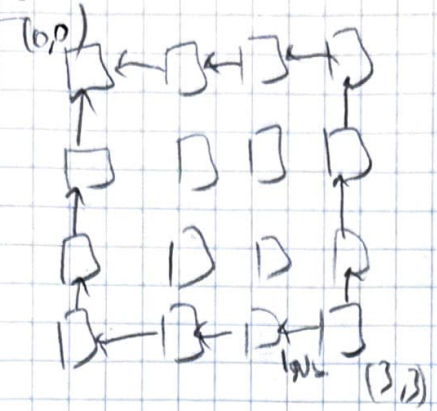
If we load from VMEM, take $\frac{258}{22}$ ie $B \geq 11$

05) TPU v5e 4x4 slice.

↳ 1 core, 1 chip. 2 hops of 4 chips/host

We want to send from TPU {0,0} to TPU {3,3}. Per hop latency = 1µs

No wrap around.



BW/link DCI = 4.5e10

total bytes = 2 × 128 × 8 × 8192

= 16,777,216 bytes

t_{stream} = 3.72e-4 seconds

$t_{2stream}$ = 3.72e-4 / 2 = 1.66e-4 seconds

= 166e-6

$t_{first\ byte}$ = 6µs (6e-6) ✓

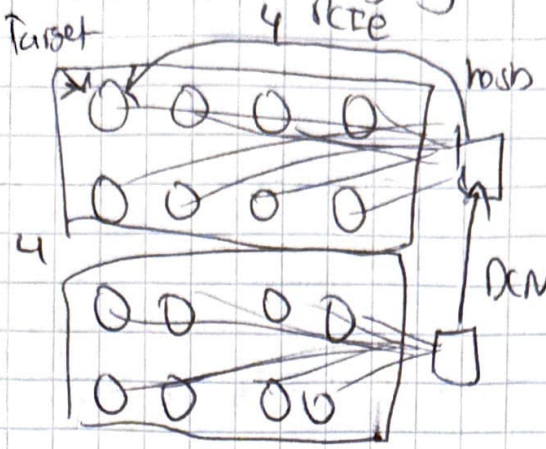
$t_{full\ byte}$ = 6µs + 166µs = 172µs ✓

Part 2-

Question 6

MatMul A: int8[128×1024, 128, 1024], sharded evenly on TPUv5e 4×4 slice, batch host DRAM. We want to copy to {0,0}, multiply by bf16[8, 128*1024].

A).



TPUv5e - Inference chip, so 1 host per 8 chips (2 trays)
ie 4×2 hosts

How to transfer data: HBM, ICI, PCIe, DCN.

HBM: Transfer to HMM
ICI: Between TPU chip and neighbours ← so have to hop to go travel long distances.
PCIe: Between CPU host and its tray(s) of chips
DCN: Between multiple CPU hosts, typically not connected by ICI.

Strategy 1

We have DCN as 3.125e9 × 8 = 2.5e10 bytes/second,
so we have to transfer 128 128

$$t_{comms\ DCN} = \frac{128 \times 1024 \times 128 \times 1024 \text{ bytes}}{2.5e10 \text{ bytes/second}} = \frac{16\text{GiB}}{5.5e10} \text{ to get to target host}$$

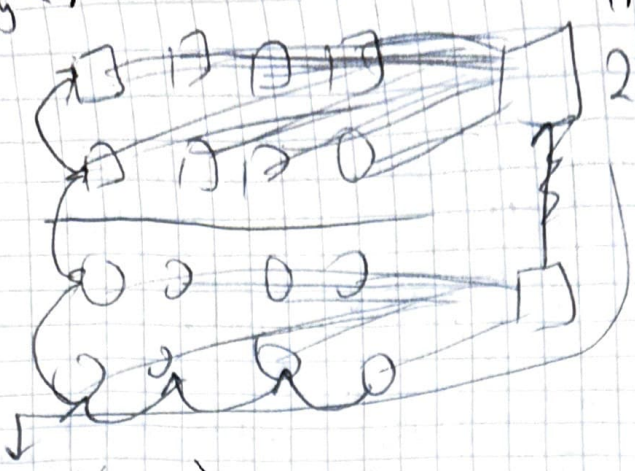
$$t_{comms\ PCIe} = \frac{128 \times 1024 \times 128 \times 16\text{GiB}}{1.6e10} \approx 1.07 \text{ seconds!!}$$

~~$t_{comms\ HBM}$ will be faster~~, so t_{comms} for HBM to VMEM is way faster than all this, so we are dominated by $t_{comms\ PCIe}$

$$t_{math} = \frac{2BDF}{3.94e14} = \frac{16\text{GiB} \cdot 2 \cdot 8 \cdot 128}{3.94e14} \approx 0.34 \text{ ms}$$

* remember to check dtype for matrix!

Strategy 2:



1. Load evenly across all PCIe
2. Hop to {0,0} over ECI.

$= \frac{16\,GiB\,(shared)}{2 \times 16e10 \times 8} \approx 0.067\,sec = 67\,ms$

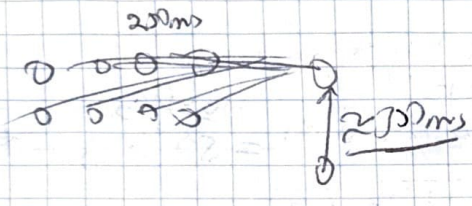
to Hop, we need 6 hops, at max, is, each transferring $\frac{16\,GiB}{16}$ bytes,

so $t_{comms}^{ECI} = \frac{16\,GiB \cdot 6}{16 \times 4 \cdot 5e10}$ bc we do not have a full ring.

$= 0.143\,sec = 143\,ms$

143ms > 67 ms, so we are bound by t_{comms} ECI.

So, the operation should take on the order of 143 ms



$t_{all} \stackrel{?}{=} max(t_{comms}, t_{mem})$

$t_{all} \leq sum(t_{comms} + t_{mem})$,

so assuming good overlapping,

143 ms is close!

Bonus:

We have $8 \cdot 128 \cdot 4.1 \cdot 15e9 = 7e12$ FLOPs/s